DPriv-Bench Category 2

Corrected Counterexample Proofs (Part2, 7 instances)

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1 Q77: Noisy numerator and noisy denominator

Original problem description

Does the following algorithm satisfy ε-differential privacy under the add/remove neighboring relationship?

Algorithm 1: Mean estimation

Input : Input scalar dataset $D := \{x_1, \ldots, x_n\}$, privacy budget ε, clipping threshold C.

Sample $\xi_1 \sim \mathrm{Lap}(2C/\varepsilon)$;

Sample $\xi_2 \sim \mathrm{Lap}(1/\varepsilon)$;

Let $\widehat{\theta} \leftarrow \dfrac{\sum_{i=1}^{n} \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$;

return $\widehat{\theta}$;

Original metadata comment

Original comment from `cate_2.json`. Under the add/remove neighboring relation, the numerator has sensitivity C and the denominator has sensitivity 1. By (sequential) composition, the resulting released estimator is $3\varepsilon/2$-DP.

Corrected proof

This composition bound does not itself disprove ε-DP: an upper bound can be loose, and the ratio is a post-processing of the two noisy statistics. A fixed event of the released ratio is required.

We use $\mathrm{Lap}(b)$ for the distribution with density

$$f_b(x) = \frac{1}{2b} e^{-|x|/b}.$$

If $X \sim \mathrm{Lap}(b)$ and $Y \sim \mathrm{Lap}(d)$ are independent, with $b \neq d$, then for $t \geq 0$,

$$T_{b,d}(t) := \Pr[X + Y > t] = \frac{b^2 e^{-t/b} - d^2 e^{-t/d}}{2(b^2 - d^2)}. \tag{1}$$

By symmetry, $\Pr[X + Y < -t] = T_{b,d}(t)$.

Set $C = \varepsilon = 1$ and take the add/remove neighbors

$$D = \{\underbrace{-1, \ldots, -1}_{20 \text{ records}}\}, \qquad D' = D \cup \{+1\}.$$

Let $X \sim \mathrm{Lap}(2)$ and $Z \sim \mathrm{Lap}(1)$ be independent. The released estimates have laws

$$M(D) = \frac{-20 + X}{20 + Z}, \qquad M(D') = \frac{-19 + X}{21 + Z}.$$

Take the fixed measurable output event $E = (-\infty, -3/2)$. With $W = X + \frac{3}{2}Z$,

$$\{M(D) \in E\} \triangle \{W < -10\} \subseteq \{Z < -20\},$$

and

$$\{M(D') \in E\} \triangle \{W < -25/2\} \subseteq \{Z < -21\}.$$

The denominator is exactly zero only on a null event. Since $\frac{3}{2}Z \sim \mathrm{Lap}(3/2)$, equation (1) gives

$$T_{2,3/2}(t) = \frac{8}{7}e^{-t/2} - \frac{9}{14}e^{-2t/3}.$$

Consequently,

$$\Pr[M(D) \in E] \geq T_{2,3/2}(10) - \frac{1}{2}e^{-20} = 0.006882388096088\ldots,$$
$$\Pr[M(D') \in E] \leq T_{2,3/2}(25/2) + \frac{1}{2}e^{-21} = 0.002051710442833\ldots.$$

Therefore,

$$\Pr[M(D) \in E] - e\Pr[M(D') \in E] \geq 0.001305260882076\ldots > 0.$$

Verdict: **The released ratio is not 1-DP.**

2 Q81: The same ratio with larger count noise

Original problem description

Does the following algorithm satisfy $\varepsilon/2$-differential privacy under the add/remove neighboring relationship?

Algorithm 2: Mean estimation

Input : Input scalar dataset $D := \{x_1, \ldots, x_n\}$, privacy budget ε, clipping threshold C.
Sample $\xi_1 \sim \mathrm{Lap}(2C/\varepsilon)$;
Sample $\xi_2 \sim \mathrm{Lap}(2/\varepsilon)$;
Let $\widehat{\theta} \leftarrow \dfrac{\sum_{i=1}^{n} \min\{\max\{-C, x_i\}, C\} + \xi_1}{n + \xi_2}$;
return $\widehat{\theta}$;

Original metadata comment

Original comment from `cate_2.json`. Under the add/remove neighboring relation, the numerator has sensitivity C and the denominator has sensitivity 1. By (sequential) composition, the resulting released estimator is ε-DP.

Corrected proof

As in Q77, the composition upper bound does not prove that the stronger $\varepsilon/2$ guarantee fails. We now give a direct event on the released ratio.

Set $C = \varepsilon = 1$ and use the same neighboring pair

$$D = \{\underbrace{-1, \ldots, -1}_{20 \text{ records}}\}, \qquad D' = D \cup \{+1\}.$$

This time let $X, Z \overset{\text{iid}}{\sim} \text{Lap}(2)$. The output laws are

$$M(D) = \frac{-20 + X}{20 + Z}, \qquad M(D') = \frac{-19 + X}{21 + Z}.$$

Take the fixed output event $E = (-1/2, \infty)$. With $W = X + \frac{1}{2}Z$,

$$\{M(D) \in E\} \triangle \{W > 10\} \subseteq \{Z < -20\},$$

whereas

$$\{M(D') \in E\} \triangle \{W > 17/2\} \subseteq \{Z < -21\}.$$

Again, zero denominators form a null event. Since $\frac{1}{2}Z \sim \text{Lap}(1)$,

$$T_{2,1}(t) = \frac{2}{3}e^{-t/2} - \frac{1}{6}e^{-t}.$$

It follows that

$$\Pr[M(D) \in E] \le T_{2,1}(10) + \frac{1}{2}e^{-10} = 0.004507097975978\ldots,$$
$$\Pr[M(D') \in E] \ge T_{2,1}(17/2) - \frac{1}{2}e^{-21/2} = 0.009461809653156\ldots.$$

Hence

$$\Pr[M(D') \in E] - e^{1/2}\Pr[M(D) \in E] \ge 0.002030861351032\ldots > 0.$$

Verdict: **The released ratio is not $1/2$-DP.**

3 Q84: Clipped ratio under replace-one adjacency

Original problem description

Does the following algorithm satisfy ε-differential privacy under the replace-one neighboring relationship?

Algorithm 3: Mean estimation

Input : Dataset $\mathcal{D} \subset [l, u]$, privacy budget $\varepsilon > 0$.
Output Noisy mean estimate $\widetilde{m}$.
:
Set $\Delta \leftarrow u - l$
Let $\mathcal{X} \leftarrow \{(x - l)/\Delta : x \in \mathcal{D}\}$
$a \leftarrow \sum_{x \in \mathcal{X}} x$, $b \leftarrow \sum_{x \in \mathcal{X}} (1 - x)$
Draw $(\eta_1, \eta_2) \sim \xi_1 \times \xi_2$, where ξ_1 and ξ_2 are i.i.d. $\mathrm{Lap}(1/\varepsilon)$
$\widehat{a} \leftarrow a + \eta_1$, $\widehat{b} \leftarrow b + \eta_2$
$\rho \leftarrow \min\{\max\{\widehat{a}/(\widehat{a} + \widehat{b}), 0\}, 1\}$
$\widetilde{m} \leftarrow l + \Delta\rho$
return $\widetilde{m}$

Original metadata comment

Original comment from `cate_2.json`. The algorithm appears as Kulesza et al. (2024, Algorithm 3), and its privacy guarantee is proved in Kulesza et al. (2024, Theorem 1) under the add/remove neighboring relation. Since the statement here asks for a guarantee under the replace-one neighboring relation, the noise scale should be doubled.

Corrected proof

The comment explains why an existing add/remove theorem is inapplicable, but inapplicability is not a proof that the displayed replace-one mechanism fails. The following exact probability calculation supplies the missing strict witness.

Set $[l, u] = [0, 1]$ and $\varepsilon = 1$, and take the replace-one neighbors

$$D_1 = \{1\}, \qquad D_0 = \{0\}.$$

Let $\eta_1, \eta_2 \overset{\text{iid}}{\sim} \mathrm{Lap}(1)$ and write

$$U = a + \eta_1, \qquad V = b + \eta_2, \qquad R = \frac{U}{U + V}, \qquad \rho = \mathrm{clip}(R, [0, 1]).$$

For D_1, $(a, b) = (1, 0)$; for D_0, $(a, b) = (0, 1)$. Since $l = 0$ and $u = 1$, the final released estimate is ρ.

Take the fixed open output event $E = (5/6, 1)$. Clipping cannot map a point outside $(0, 1)$ into this open interval, so $\{\rho \in E\} = \{R \in E\}$. Outside the null event $\{U + V = 0\}$,

$$\{R \in E\} = \{V > 0,\ U > 5V\} \cup \{V < 0,\ U < 5V\}.$$

Conditioning on V and splitting the Laplace integrals at their breakpoints yields

$$\begin{aligned} p_1 := \Pr[M(D_1) \in E] &= \frac{1}{2}(1 - e^{-1/5}) - \frac{e^{-1}(e^{4/5} - 1)}{16} + \frac{e^{-1/5}}{24} + \frac{e^{-1}}{24} \\ &= 0.111898507893910\ldots, \\ p_0 := \Pr[M(D_0) \in E] &= \frac{e^{-1}(1 - e^{-4})}{16} + \frac{e^{-5}}{24} + \frac{e^{-1}}{24} \\ &= 0.038180401226211\ldots. \end{aligned}$$

Therefore,

$$p_1 - e p_0 = 0.008113417037425\ldots > 0.$$

Verdict: **The final clipped mean is not 1-DP under replace-one adjacency.**

4 Q104: Parallel composition with a free privacy index

Original problem description

Consider the following procedure: $\mathcal{M} := \{\mathcal{M}_i(X_i)\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$-differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\varepsilon_i, \delta_i)$-differential privacy?

Original metadata comment

Original comment from `cate_2.json`. Under the assumptions that the subsets are disjoint and the partition is data-independent, the combined mechanism follows the parallel composition and post-processing property of differential privacy, this implies that $\mathcal{M}$ satisfies $(\max_{i\in[n]} \varepsilon_i, \max_{i\in[n]} \delta_i)$-differential privacy, rather than the stronger $(\varepsilon_i, \delta_i)$-DP guarantee stated.

Corrected proof

The comment identifies the usual parallel-composition bound, but the question's displayed pair contains a free index i. A rigorous resolution must first quantify that index and, for a No answer, exhibit a mechanism that actually violates the selected pair.

Literal-text issue. Standard differential privacy assigns one numerical privacy pair to the whole release. The expression $(\varepsilon_i, \delta_i)$ is therefore not a well-formed whole-mechanism claim until i is fixed or quantified. The counterexample below addresses either the reading "for every i" or the claim for a specified index whose budget does not dominate the other component budgets.

We first define a common primitive used for Q104–Q106. Let $\lambda = \log 3$ and let binary randomized

response be

$$\mathrm{RR}_\lambda(b) = \begin{cases} b, & \text{with probability } \dfrac{e^\lambda}{1+e^\lambda} = \dfrac{3}{4}, \\ 1-b, & \text{with probability } \dfrac{1}{1+e^\lambda} = \dfrac{1}{4}. \end{cases}$$

For the replace-one neighbors $D_0 = \{0\}$ and $D_1 = \{1\}$, this mechanism is exact pure λ-DP. Its bound is attained on the fixed event $\{1\}$.

Fix the queried index to be 1. Take two fixed, data-independent disjoint blocks

$$X_1 = \emptyset, \qquad X_2 = X.$$

Let $\mathcal{M}_1$ be constant and let $\mathcal{M}_2 = \mathrm{RR}_\lambda$. Declare

$$(\varepsilon_1, \delta_1) = (0, 0), \qquad (\varepsilon_2, \delta_2) = (\lambda, 0).$$

In the joint output space, use the same fixed event for both neighbors,

$$A = \{(y_1, y_2) : y_2 = 1\}.$$

Then

$$\Pr[\mathcal{M}(D_1) \in A] = \frac{3}{4}, \qquad \Pr[\mathcal{M}(D_0) \in A] = \frac{1}{4}.$$

The claimed $(\varepsilon_1, \delta_1) = (0, 0)$ inequality would require

$$\frac{3}{4} \le e^0 \frac{1}{4} + 0 = \frac{1}{4},$$

which is false.

The correct general guarantee is

$$\mathcal{M} \text{ is } (\varepsilon_\star, \delta_\star)\text{-DP}, \qquad \varepsilon_\star = \max_j \varepsilon_j, \quad \delta_\star = \max_j \delta_j.$$

For a specified index k, this bound implies $(\varepsilon_k, \delta_k)$-DP when $\varepsilon_k \ge \varepsilon_\star$ and $\delta_k \ge \delta_\star$, for example when all component budgets are equal or k attains both maxima.

Verdict: **Under the "for every i" reading, the fixed event above strictly refutes the claim. If the intended pair is the coordinatewise maximum, the statement is true by parallel composition.**

5 Q105: Repeated access to parity blocks

Original problem description

Consider the following mechanism: $\mathcal{M}(X) := \{\mathcal{M}_i(X_{i \bmod 2})\}_{i=1}^n$, where each subset X_i is disjoint from the others, i.e., $X_i \cap X_j = \emptyset$ for all $i \neq j$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$-differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i\in[n]} \varepsilon_i, \max_{i\in[n]} \delta_i)$-differential privacy?

Original metadata comment

Original comment from `cate_2.json`. The overall mechanism can be written as

$$\mathcal{M} := \left\{\{\mathcal{M}_i(X_0)\}_{i\in A}, \{\mathcal{M}_i(X_1)\}_{i\in B}\right\},$$

where $A := \{i \in [n] : i \text{ is even}\}$ and $B := \{i \in [n] : i \text{ is odd}\}$. Consequently, $\mathcal{M}$ satisfies

$$\left(\max\left\{\sum_{i\in A}\varepsilon_i, \sum_{i\in B}\varepsilon_i\right\}, \max\left\{\sum_{i\in A}\delta_i, \sum_{i\in B}\delta_i\right\}\right)\text{-DP}.$$

Corrected proof

The comment gives the appropriate basic-composition upper bound after repairing the indexing, but it does not by itself show that the smaller max bound is false. The construction below makes both the repair and the violating event explicit.

Literal-text issue. With $i \in \{1, \dots, n\}$, the expression $X_{i \bmod 2}$ references X_0 for every even i, although X_0 is never defined. We adopt the natural repaired reading $X = X_0 \dot\cup X_1$, with every even mechanism run on X_0 and every odd mechanism run on X_1. Equivalently, one may reindex these two blocks as X_1, X_2.

Take $n = 3$, let

$$X_0 = \emptyset, \qquad X_1 = X,$$

and define

$$\mathcal{M}_1 = \mathcal{M}_3 = \mathrm{RR}_\lambda, \qquad \mathcal{M}_2 \text{ constant},$$

where the two randomized-response calls use independent random coins. Thus

$$(\varepsilon_1, \delta_1) = (\varepsilon_3, \delta_3) = (\lambda, 0), \qquad (\varepsilon_2, \delta_2) = (0, 0),$$

so the claimed max pair is $(\lambda, 0)$.

For the fixed neighboring pair $D_0 = \{0\}$ and $D_1 = \{1\}$, consider the fixed event

$$A = \{(y_1, y_2, y_3) : y_1 = 1 \text{ and } y_3 = 1\}.$$

Independence gives

$$\Pr[\mathcal{M}(D_1) \in A] = \left(\frac{3}{4}\right)^2 = \frac{9}{16}, \qquad \Pr[\mathcal{M}(D_0) \in A] = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$$

The claimed $(\lambda, 0)$ inequality would require

$$\frac{9}{16} \le e^{\lambda} \frac{1}{16} = \frac{3}{16},$$

a strict contradiction. The event likelihood ratio is $9 = e^{2\lambda}$.

For the repaired two-block mechanism, basic sequential composition within each parity class and parallel composition across the two blocks give

$$\varepsilon_{\text{par}} = \max_{b \in \{0,1\}} \sum_{\substack{1 \le i \le n \\ i \bmod 2 = b}} \varepsilon_i,$$
$$\delta_{\text{par}} = \max_{b \in \{0,1\}} \sum_{\substack{1 \le i \le n \\ i \bmod 2 = b}} \delta_i.$$

If $X_{i \bmod 2}$ was merely a typographical error for X_i, then ordinary parallel composition would instead make the original max claim true.

Verdict: **Under the explicit two-block repair, repeated access incurs sequential privacy loss and the claimed max guarantee fails. The unrepaired question has no mathematical truth value.**

6 Q106: Adjacent disjointness is not a partition

Original problem description

Consider the following procedure: $\mathcal{M} := \{\mathcal{M}_i(X_i)\}_{i=1}^n$, where $X_i \cap X_{i+1} = \emptyset$ for all $i \in \{1, \ldots, n-1\}$, and their union covers the entire dataset, i.e., $\bigcup_{i=1}^n X_i = X$. Furthermore, the partition is data-independent. Suppose each mechanism $\mathcal{M}_i$ satisfies $(\varepsilon_i, \delta_i)$-differential privacy. Does the combined mechanism $\mathcal{M}$ satisfy $(\max_{i \in [n]} \varepsilon_i, \max_{i \in [n]} \delta_i)$-differential privacy?

Original metadata comment

Original comment from `cate_2.json`. Consider the following counter example: let $X_i = X$ when i is odd and $X_i = \emptyset$ when i is even, then $\mathcal{M}$ satisfies $\mathcal{M}$ satisfies

$$\left(\sum_{\substack{i \in [n] \\ i \text{ is odd}}} \varepsilon_i, \sum_{\substack{i \in [n] \\ i \text{ is odd}}} \delta_i \right)\text{-DP}.$$

Corrected proof

This proposed layout is useful, but a composition upper bound is not a strict privacy violation. The corrected proof fixes a neighboring pair and one common output event.

Literal-text issue. The displayed condition only makes consecutive blocks disjoint. Nonconsecutive blocks may overlap, so $X_1, \ldots, X_n$ need not be a partition. The word "partition" conflicts with the formula. The counterexample below gives priority to the displayed condition and treats the blocks as a fixed data-independent cover.

Take $n = 3$ and use the fixed cover

$$X_1 = X_3 = X, \qquad X_2 = \emptyset.$$

It satisfies

$$X_1 \cap X_2 = X_2 \cap X_3 = \emptyset, \qquad X_1 \cup X_2 \cup X_3 = X.$$

Let

$$\mathcal{M}_1 = \mathcal{M}_3 = \mathrm{RR}_\lambda, \qquad \mathcal{M}_2 \text{ constant},$$

again with independent random coins. The declared max pair is $(\lambda, 0)$. For the fixed replace-one neighbors $D_0 = \{0\}$ and $D_1 = \{1\}$, use the same fixed event

$$A = \{(y_1, y_2, y_3) : y_1 = 1 \text{ and } y_3 = 1\}.$$

Then

$$\Pr[\mathcal{M}(D_1) \in A] = \frac{9}{16}, \qquad \Pr[\mathcal{M}(D_0) \in A] = \frac{1}{16}.$$

The claimed inequality would again require

$$\frac{9}{16} \le e^\lambda \frac{1}{16} = \frac{3}{16},$$

which is strictly false.

For a general fixed cover, let $I(u) = \{i : u \in X_i\}$ be the indices of mechanisms whose inputs contain individual u. Basic composition gives

$$\varepsilon_{\mathrm{cov}} = \max_u \sum_{i \in I(u)} \varepsilon_i, \qquad \delta_{\mathrm{cov}} = \max_u \sum_{i \in I(u)} \delta_i.$$

Consecutive disjointness only prevents $I(u)$ from containing consecutive indices; it does not force $|I(u)| \le 1$.

Verdict: Under the displayed adjacent-disjointness condition, the claimed max guarantee fails. If "partition" is instead enforced in its usual pairwise-disjoint sense, parallel composition makes the answer Yes.

7 Q136: Sparse vector technique

Original problem description

Does the following implementation of the sparse vector technique (SVT) satisfy $\varepsilon/2$-DP? (assume $c > 1$)

Algorithm: SVT

Input. A database $\mathcal{D}$, a query sequence $\{q_i\}_{i=1}^{\infty}$ each with sensitivity at most Δ, a set of thresholds $\{T_i\}_{i=1}^{\infty}$, privacy budget ε, and an answer cap $c \in \mathbb{N}$.
Procedure.
1. Split privacy budget: $\varepsilon_1 = \varepsilon_2 = \varepsilon/2$.
2. Sample a noisy reference threshold $\theta \sim \mathrm{Lap}(\Delta/\varepsilon_1)$.
3. Initialize counter $r \leftarrow 0$.
4. For each query q_i:
   (a) Draw $\eta_i \sim \mathrm{Lap}(2c\Delta/\varepsilon_2)$.
   (b) If $q_i(\mathcal{D}) + \eta_i \geq T_i + \theta$, output $\top$ and increment r.
   (c) Stop if $r \geq c$; otherwise continue.
   (d) If the condition fails, output $\perp$.

Original metadata comment

Original comment from `cate_2.json`. The privacy guarantee comes from Lyu et al. (2017, Algorithm 1), should be ε-DP by adaptive composition.

Corrected proof

The comment gives the expected upper bound but does not prove failure of $\varepsilon/2$-DP. We construct a finite transcript whose likelihood ratio is strictly larger than $e^{\varepsilon/2}$.

Control-flow interpretation. We use the standard SVT order: on failure, output $\perp$ and continue; on success, output $\top$, and stop only after c successes. Under a literal reading in which "otherwise continue" executes before the failure output, the displayed pseudocode is incomplete.

Set

$$\varepsilon = 1, \qquad \Delta = 1, \qquad c = 2.$$

The exact scales in the question are

$$\theta \sim \mathrm{Lap}(2), \qquad \eta_i \overset{\text{iid}}{\sim} \mathrm{Lap}(8).$$

Let the neighboring databases be bits $D = 0$ and $D' = 1$. Define

$$q_i(x) = x \quad (i = 1, 2, 3), \qquad q_i(x) = 1 - x \quad (i = 4, 5),$$

and set $T_i = 0$. Every query has sensitivity 1. For $i \geq 6$, set $q_i(x) = 0$ as a valid infinite extension; the event below stops at query 5, so later queries do not affect it.

Consider the complete transcript

$$E = (\bot, \bot, \bot, \top, \top).$$

The second $\top$ reaches the cap $c = 2$, so the mechanism stops legally. Let F_8 be the CDF of Lap(8), let $\overline{F}_8 = 1 - F_8$, and let

$$f_2(t) = \frac{1}{4} e^{-|t|/2}.$$

Conditioning on the shared noisy threshold $\theta = t$ gives

$$P_D(E) = \int_{\mathbb{R}} f_2(t) F_8(t)^3 \overline{F}_8(t-1)^2 \, dt,$$
$$P_{D'}(E) = \int_{\mathbb{R}} f_2(t) F_8(t-1)^3 \overline{F}_8(t)^2 \, dt.$$

Let $r = e^{1/8}$. Splitting each integral at $t = 0$ and $t = 1$ yields

$$P_D(E) = \frac{3}{20} - \frac{33}{160r} + \frac{127}{1440r^2} - \frac{1}{140r^5} + \frac{1}{320r^6} - \frac{1}{2520r^7},$$
$$P_{D'}(E) = \frac{65}{2016r^3} - \frac{19}{4032r^6}.$$

Numerically,

$$P_D(E) = 0.0341583443607506\ldots, \qquad P_{D'}(E) = 0.0199336909079234\ldots.$$

Hence

$$\frac{P_D(E)}{P_{D'}(E)} = 1.71359857632653\ldots > e^{1/2} = 1.64872127070013\ldots,$$

or equivalently,

$$P_D(E) - e^{1/2} P_{D'}(E) = 0.00129324415729553\ldots > 0.$$

The same likelihood ratio is below $e^{\varepsilon} = e$, consistently with the standard ε-DP upper bound.

Verdict: **The displayed SVT mechanism, under the standard output order, is not $\varepsilon/2$-DP.**

8 Independent numerical checks

All decimals used in the corrected proofs were recomputed from their displayed closed forms. The table records the quantity that lies beyond the claimed privacy boundary.

Item	Checked quantity	Strict margin
Q77	$P_D(E) - eP_{D'}(E)$	0.0013052609
Q81	$P_{D'}(E) - e^{1/2}P_D(E)$	0.0020308614
Q84	$p_1 - ep_0$	0.0081134170
Q104	$3/4 - 1/4$ under the claimed $(0, 0)$ pair	0.5
Q105	$9/16 - 3/16$	0.375
Q106	$9/16 - 3/16$	0.375
Q136	$P_D(E) - e^{1/2}P_{D'}(E)$	0.0012932442

References

B. Balle and Y.-X. Wang. *Improving the Gaussian Mechanism for Differential Privacy: Analytical Calibration and Optimal Denoising*. ICML, 2018.

J. Dong, A. Roth, and W. J. Su. *Gaussian Differential Privacy*. Journal of the Royal Statistical Society: Series B, 2022.

A. Kulesza et al. *Mean Estimation in the Add-Remove Model of Differential Privacy*. ICML, 2024.

M. Lyu, D. Su, and N. Li. *Understanding the Sparse Vector Technique for Differential Privacy*. PVLDB, 2017.

I. Mironov. *Renyi Differential Privacy*. 2017.

N. Papernot and T. Steinke. *Hyperparameter Tuning with Renyi Differential Privacy*. ICLR, 2022.